

PROJECT REDWOOD

Take-Private Underwriting, Reverse LBO & Probabilistic Bid Strategy

A synthetic Investment Banking / Private Equity case study built around one decision: **what is the highest price a sponsor can pay while preserving return hurdles, debt capacity and downside protection?**

Entry EV / EBITDA	8.5x	Gross IRR	22.3%
Gross MOIC	2.74x	25% IRR bid ceiling	8.1x
Bid gap	(0.4x) / (\$24.5mm)	Initial leverage	5.0x

IC readout: the base case clears the 2.5x MOIC hurdle but falls short of the 25% IRR hurdle. The deal therefore requires a lower bid, stronger operating performance, more deleveraging, or a more aggressive exit assumption.

	A	B	C	D	E	F	G	H
1	Investment Committee Snapshot							
2	Synthetic Project Redwood decision-oriented summary							
3								
4	Key Transaction Metrics							
5	Entry EV / EBITDA		Enterprise Value		Sponsor Equity		Initial Gross Leverage	
6	8.5x		\$524.9		\$223.4		5.0x	
7								
8	Exit EV / EBITDA		Gross MOIC		Gross IRR		Max Entry Multiple @ Hurdles	
9	9.5x		2.7x		22.3%		8.1x	
10								
11	Bid Gap		Min Interest Coverage					
12	(0.4x)		2.7x					
13								
14	Investment Committee Readout							
15	Base case produces 2.74x MOIC and 22.3% gross IRR. The 25% IRR hurdle implies a maximum entry multiple of 8.1x, creating a bid gap of 0.4x versus the current 8.5x asking valuation. Decision framing: the project is deliberately structured so that the sponsor must choose between (i) lowering the bid, (ii) underwriting stronger operating performance, (iii) accepting a lower return hurdle, or (iv) relying on more aggressive exit assumptions. This makes the model a bid-discipline exercise rather than a mechanical LBO template.							
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1. Transaction Structure & Financing

The model translates an 8.5x LTM EBITDA purchase price into sources and uses, management rollover and sponsor equity. Debt is sized at 5.0x gross leverage through a Term Loan B and second-lien tranche, with minimum cash and transaction / financing fees funded at close.

	A	B	C	D	E	F
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4	Transaction Valuation					
5	LTM Revenue	\$325.0				
6	LTM EBITDA Margin	19.0%				
7	LTM EBITDA	\$61.8				
8	Entry EV / EBITDA	8.5x				
9	Enterprise Value	\$524.9				
10	Equity Purchase Price	\$479.9				
11						
12	Uses					
13	Equity Purchase Price	\$479.9		Term Loan B		\$247.0
14	Refinance Existing Net Debt	\$45.0		Second-Lien Debt		\$61.8
15	Minimum Cash	\$10.0		Management Rollover		\$19.4
16	Transaction Fees	\$10.5		Sponsor Equity		\$223.4
17	Financing Fees	\$6.2		Total Sources		\$551.5
18	Total Uses	\$551.5		Sources - Uses Check		-
19						
20						
21	Gross Debt	\$308.8				
22	Cash	\$10.0				
23	Net Debt	\$298.8				
24	Gross Leverage	5.0x				
25	Net Leverage	4.8x				
26	Sponsor % of Equity Funding	92.0%				
27	Management % of Equity Funding	8.0%				

Credit discipline: the debt schedule uses 1% mandatory TLB amortisation, a 75% cash sweep, an RCF liquidity backstop and a 2.0x minimum interest-coverage guardrail. Base-case minimum interest coverage is 2.7x.

2. Reverse LBO & Bid Discipline

Instead of accepting the seller's asking price and reporting whatever return falls out, the reverse LBO works backward from sponsor exit proceeds and the 25% IRR / 2.5x MOIC hurdles to determine the maximum supportable entry valuation.

	A	B	C	D	E	F
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4	Base-Case Bid Ceiling					
5	Sponsor Exit Proceeds	\$611.5				
6	Target IRR	25.0%				
7	Target MOIC	2.5x				
8	Maximum Sponsor Equity - IRR	\$200.4				
9	Maximum Sponsor Equity - MOIC	\$244.6				
10	Binding Sponsor Equity	\$200.4				
11	Maximum Total Equity Funding	\$217.8				
12	Debt Sources	\$308.8				
13	Minimum Cash	\$10.0				
14	Transaction Fees % EV	0.02				
15	Financing Fees	\$6.2				
16	Maximum Enterprise Value	\$500.4				
17	Maximum Entry EV / EBITDA	8.1x				
18	Current Entry EV / EBITDA	8.5x				
19	Bid Gap	(0.4x)				
20	Bid Gap (\$mm EV)	(\$24.5)				
21						
22						
23						
24	Exit Multiple / Target IRR	20.0%	22.5%	25.0%	27.5%	30.0%
25	8.0x	8.0x	7.7x	7.4x	7.2x	6.9x
26	9.0x	8.6x	8.2x	7.9x	7.6x	7.3x
27	9.5x	8.9x	8.5x	8.1x	7.8x	7.5x
28	10.0x	9.2x	8.7x	8.3x	8.0x	7.7x
29	11.0x	9.7x	9.2x	8.8x	8.4x	8.1x

The binding constraint is the IRR hurdle, producing a base-case maximum entry multiple of 8.1x versus an 8.5x ask. That equates to a \$24.5mm EV re-pricing requirement.

3. Hurdle Solver - What Must Improve at the Current Price?

The hurdle solver frames three independent ways the current 8.5x bid could still reach a 25% gross IRR, holding the other assumptions constant.

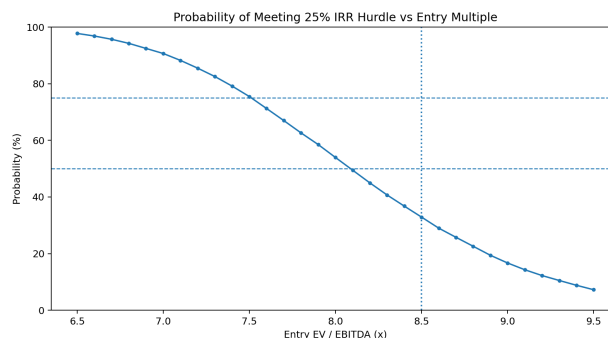
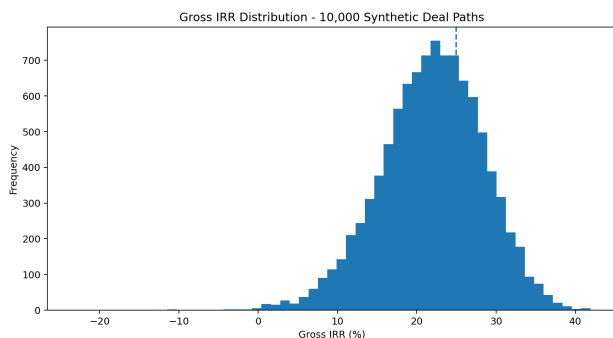
	A	B	C	D	E	F
1	Bid Requirements & Hurdle Solver					
2	Synthetic Project Redwood what must be true to justify the seller's 8.5x asking valuation?					
3						
4	Current Bid Gap					
5	Metric	Base Case / Ask	Hurdle / Ceiling	Gap		
6	Entry EV / EBITDA	8.5x	8.1x	(0.4x)		
7	Enterprise Value	\$524.9	\$500.4	(\$24.5)		
8	Gross IRR	22.3%	25.0%	(2.7%)		
9	Gross MOIC	2.74x	2.50x	0.24x		
10	Decision	Current asking price	Base-case max bid	RE-PRICE / IMPROVE CASE		
11						
12	If the Sponsor Pays 8.5x, What Must Improve to Hit 25% IRR?					
13	Value Driver	Base Case	Required @ 25% IRR	Change vs Base	Hurdle Solver Helpers	Value
14	Exit EBITDA	\$87.9	\$95.9	9.1%	Required Sponsor Exit Proceeds	\$681.7
15	Exit EBITDA CAGR	7.3%	9.2%	1.9%	Required Total Exit Equity	\$741.0
16	Exit EV / EBITDA	9.5x	10.4x	0.9x	Required Exit EV @ Base Debt	\$911.5
17	Exit Net Debt	\$170.5	\$94.2	(\$76.3)	Required Exit EV @ Base EBITDA	\$911.5
18	Net Leverage at Exit	1.9x	1.1x	(0.9x)	Required Exit Net Debt @ Base EV	\$94.2
19						
20	Decision Frame					
21	At the current 8.5x ask, the base case generates 22.3% gross IRR. To reach the 25% hurdle without repricing, the sponsor would need approximately 9.1% more exit EBITDA, a 0.9x higher exit multiple, or \$76.3mm of additional debt paydown versus the base case.					
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Route	Base	Required	Change
Exit EBITDA	\$87.9mm	\$95.9mm	+9.1%
Exit multiple	9.5x	10.4x	+0.9x
Exit net debt	\$170.5mm	\$94.2mm	(\$76.3mm)

4. Probabilistic Underwriting Layer

The companion Python engine runs 10,000 synthetic deal paths with correlated shocks to revenue growth, EBITDA margin, base rates and exit valuation. Each path passes through the same operating and debt waterfall as the deterministic model.

Median IRR	22.2%	P(IRR >= 25%)	32.9%
5th / 95th IRR	10.8% / 32.0%	P(coverage < 2.0x)	0.6%
50% bid ceiling	8.0x	75% bid ceiling	7.5x



At the seller's 8.5x ask, the 25% IRR hurdle is met in only about one-third of simulated paths. A 50% probability threshold implies roughly an 8.0x bid ceiling; a 75% threshold implies roughly 7.5x.

5. Investment Committee Framing

Project Redwood is designed as a **decision exercise**, not a model that is engineered to produce a passing return. The sponsor has four credible choices:

1	Re-price the bid	Move toward the 8.1x deterministic ceiling or lower.
2	Underwrite stronger operations	Require approximately 9% more exit EBITDA than the current base case.
3	Rely on valuation / deleveraging	Accept a 10.4x exit multiple or materially stronger debt paydown.
4	Accept lower expected returns	Proceed at 8.5x with a return profile below the 25% hurdle in the base case.

	A	B	C	D	E	F	G
1	Probabilistic Underwriting - 10,000 Synthetic Deal Paths						
2	Companion Python risk engine fixed seed 77 synthetic educational assumptions						
3							
4	Key Risk Outputs						
5	Metric	Result	Metric	Result	Metric	Result	
6	Median Gross IRR	22.2% P(IRR >= 25%)		32.9% P(Both Hurdles)		32.9%	
7	5th %ile Gross IRR	10.8% 95th %ile Gross IRR		32.0% Median Gross MOIC		2.7x	
8	P(Coverage < 2.0x)	0.6% P(RCF Draw)		0.1% Median Exit Net Debt		\$170.8	
9	Risk-Adjusted Bid Ceiling @ 50%	8.0x Risk-Adjusted Bid Ceiling @ 75%		7.5x Deterministic Max Bid		8.1x	
10	Current Asking Multiple	8.5x Base-Case IRR		22.3% Base-Case MOIC		2.7x	
11							
12	What Drives Gross IRR?						
13	Driver	Spearman Corr.		Entry Multiple	P(IRR >=25%)	Median IRR	5th %ile IRR
14	Exit Net Debt	(0.96)		7.0x	90.7%	34.9%	22.3%
15	Average Revenue Growth	0.90		7.5x	75.5%	29.8%	17.7%
16	Exit EBITDA Margin	0.84		8.0x	54.0%	25.6%	13.9%
17	Exit EV / EBITDA	0.80		8.1x	49.5%	24.9%	13.3%
18	Average Base Rate	(0.50)		8.5x	32.9%	22.2%	10.8%
19				9.0x	16.7%	19.3%	8.1%
20							
21	Underwriting Interpretation						
22							
23	The deterministic reverse LBO sets a base-case ceiling of 8.1x EV / EBITDA. The Monte Carlo layer asks a different question: how much can the sponsor pay if it wants a given probability of clearing the 25% IRR hurdle? Under the synthetic distribution used here, the ceiling is approximately 8.0x for a 50% probability and 7.5x for a 75% probability. The current 8.5x asking price clears the 25% IRR hurdle in only about one-third of simulated paths.						
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27	Model note: this risk layer is deliberately synthetic and is not calibrated to a live transaction or market data. Its purpose is to demonstrate probabilistic deal underwriting, bid discipline and the interaction between operations, financing and valuation.						
28							

Model integrity: the Python engine reconciles the deterministic Excel case and includes 8 automated tests covering base-case returns, debt, reproducibility and bid-probability behaviour.

All company data, transaction terms and probability distributions are fictional and created solely to demonstrate modelling, underwriting and deal-team decision mechanics.